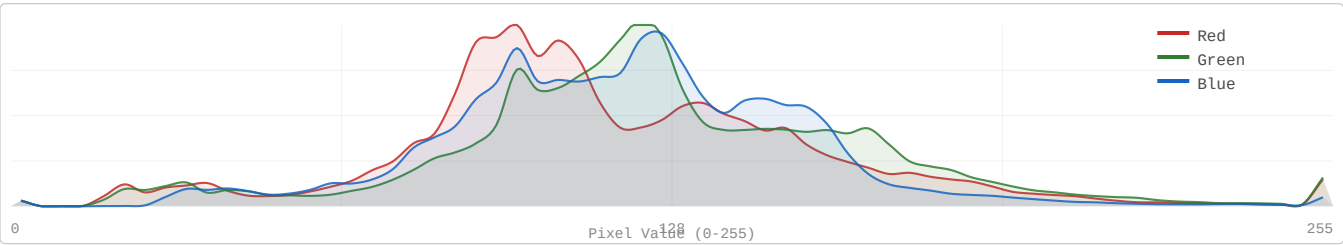


Decorrelation Gap: KODIM10

512x768 | 24-bit RGB | PNG (lossless)

Flower Garden and Wooden Fence | Strongly one-dimensional (PC1 = 96.12%)

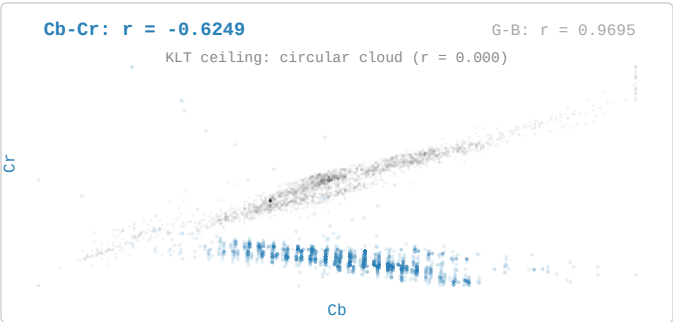
1. RGB Channel Distribution



2. Inter-Channel Correlation: RGB vs YCbCr



3. Cb vs Cr Scatter with G-B background overlay



4. Pairwise Decorrelation Gap

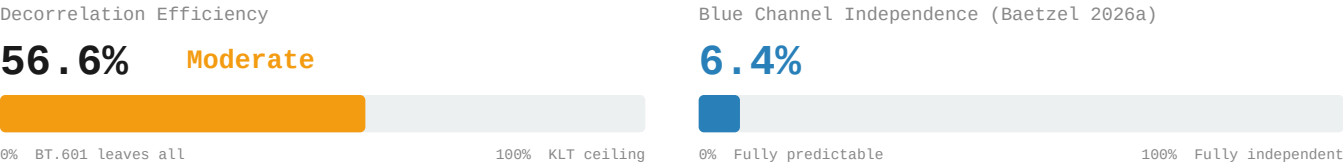
Channel Pair	RGB	YCbCr (BT.601)	Residual r
R-G → Y-Cb	0.955452	-0.590750	0.5907
R-B → Y-Cr	0.890182	-0.005614	0.0056
G-B → Cb-Cr	0.969513	-0.624874	0.6249
Avg r	0.938382	0.407080	0.4071

Decorrelation Gap: KODIM10

512x768 | 24-bit RGB | PNG (lossless)

Flower Garden and Wooden Fence | Strongly one-dimensional (PC1 = 96.12%)

5. Decorrelation Efficiency & Blue Independence



6. 4:2:0 Chroma Subsampling – Information Cost

Channel	PSNR (dB)
R	45.85 dB
G	49.56 dB
B	43.55 dB
Overall	45.67 dB

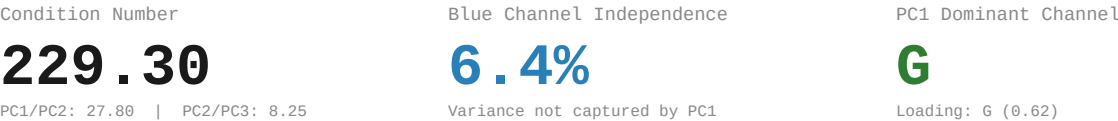
Method: BT.601 RGB→YCbCr | 2x box downsample Cb,Cr | nearest upsample | BT.601 inverse

7. PCA Baseline Reference (Baetzel 2026a)
Eigendecomposition



Eigenvector Loadings				
	R	G	B	Eigenvalue
PC1	-0.6010	-0.6226	-0.5011	4279.40
PC2	-0.7275	0.1666	0.6656	153.95
PC3	0.3309	-0.7646	0.5530	18.66

8. PCA Derived Metrics



9. Redundancy Profile

Classification:	Strongly one-dimensional CN: 229.30 PC1: 96.1%
RGB Correlation:	Avg r = 0.938382 R-G: 0.9555 R-B: 0.8902 G-B: 0.9695
YCbCr Residual:	Avg r = 0.407080 Y-Cb: -0.5907 Y-Cr: -0.0056 Cb-Cr: -0.6249
Efficiency:	56.6% of KLT ceiling Gap: 0.4071 avg r remaining
Blue Channel:	6.4% independent 4:2:0 PSNR: 45.67 dB

References

[1] Baetzel, J. (2026a). "Statistical Characterization of Inter-Channel Redundancy Structure"

[2] Wallace, G.K. "The JPEG still picture compression standard." IEEE Trans. CE 38(1), 1992.

[3] ITU-R Recommendation BT.601: Studio Encoding Parameters, 1982.

[4] Watanabe, S. "Karhunen-Loeve Expansion and Factor Analysis," pp. 635-660, 1965.